

SixEyes – Follower PCB Beta (Rev 2)

Technical Design Specification & KiCad Layout Manual

Target System: SixEyes NodeMesh VLA – Node0 (Orchestrator)
Module: ESP32-S3-WROOM-1-N16R8 (16 MB Flash, 8 MB Octal PSRAM)
Form Factor: 60 mm × 60 mm Custom 4-Layer PCB
Placement Rule: Single-Sided SMT (Top Layer Only)

Rev 2 changes: LDO upgraded to 1 A; bulk caps changed to low-ESR electrolytic; TMC2209 DIAG/StallGuard pins routed; full connector edge map added; CH340K decoupling added; 5 V buck load budget revised for Node1 (ESP32-C6 offboard); octal-PSRAM locked GPIO exclusions enforced.

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1 System Overview & Architecture

The Follower PCB Beta (Rev 2) is the physical carrier board for **Node0** of the SixEyes NodeMesh VLA system. Node0 is the real-time orchestrator providing:

- **400 Hz deterministic control** – trajectory profiling for four stepper joints and three servo joints simultaneously.
- **2-link planar inverse kinematics** – upper arm / forearm link length = 140 mm each.
- **Data aggregation** – synchronous merging of joint-state frames from Node1 (ESP32-C6 leader board, offboard, connected via hardware UART at 921 600 baud) and vision feature vectors from Node2 / Node3 (ESP32-CAM modules, wireless ESP-NOW on channel 6).
- **Full TMC2209 feature utilisation** – StallGuard2 sensorless homing, CoolStep adaptive current scaling, SpreadCycle / StealthChop2 mode switching, and per-driver UART configuration via PDN_UART single-wire bus.
- **Telemetry logging** – MicroSD circular-buffer experience packet logging (≈ 300 B/packet, FAT32).

1.1 Module Selection Note – Octal PSRAM Variant

The **ESP32-S3-WROOM-1-N16R8** is the required module variant (16 MB Flash, 8 MB Octal PSRAM). The octal-PSRAM interface permanently commits the following GPIO lines to internal silicon use. **These pins must not be routed to any external pad, via, connector, or pull resistor on the PCB:**

GPIO	Internal Function (N16R8 variant)
GPIO33	Flash D4
GPIO34	Flash D5
GPIO35	Octal PSRAM D6 / WP
GPIO36	Octal PSRAM D7 / HOLD
GPIO37	Octal PSRAM CLK

GPIO26 is additionally used by the internal octal flash clock and must not be routed.

1.2 Power Topology

24 V DC Input $\rightarrow$ 4 $\times$ TMC2209 V_M rails (direct, fused)
 $\rightarrow$ 6.6 V sync buck (TPS54540B) $\rightarrow$ 3 $\times$ MG996R/MG995 servos
 $\rightarrow$ 5.0 V async buck (LMR14030S) $\rightarrow$ 2 $\times$ ESP32-CAM headers (Node2, Node3)
 $\rightarrow$ ESP32-C6 offboard header (Node1 leader board)
 $\rightarrow$ 3.3 V LDO (MIC5219-3.3) $\rightarrow$ ESP32-S3 core logic, TMC2209 V_{IO}

2 Power Delivery Network (PDN) & Domain Architecture

2.1 24 V Motor Rail

The four TMC2209 V_M pins are fed directly from the 24 V input bus. Each driver requires the following filtering placed within 2 mm of the IC body:

- 100 μF / 35 V **low-ESR SMD electrolytic** (Panasonic FR or Nichicon UWX series, case size D / 7.3 mm diameter) – bulk transient reservoir per driver. **Do not substitute tantalum; polarity reversal or inrush spikes from motor back-EMF cause catastrophic tantalum failure.**
- 100 nF X7R ceramic, 50 V (0603) – high-frequency bypass, trace length to pin ≤ 1 mm.

2.2 6.6 V Servo Domain

Regulator: TPS54540B – 5.5 A synchronous buck, HSOP-8-EP package. Readily available, comprehensive datasheet, budget-friendly.

Loads: 3 $\times$ MG996R/MG995 digital servos (Wrist Pitch, Wrist Yaw, Gripper).

Continuous / peak: 5.0 A / 7.0 A stall.

Inductor: Shielded iron-powder / metal composite, $L = 4.7 \mu\text{H}$, $I_{\text{sat}} > 9.0$ A, footprint $\geq 10 \text{ mm} \times 10 \text{ mm}$ (e.g. Bourns SRR1260 or Sumida CDEP134).

Output capacitor: $\geq 1000 \mu\text{F}$ low-ESR SMD electrolytic (2 $\times$ 470 μF / 16 V in parallel acceptable) plus 10 μF ceramic bank at servo headers.

Feedback resistors (1% tolerance, 0603):

$$V_{\text{out}} = V_{\text{ref}} \times \left(1 + \frac{R_{\text{top}}}{R_{\text{bot}}}\right) = 0.8 \text{ V} \times \left(1 + \frac{102 \text{ k}\Omega}{14 \text{ k}\Omega}\right) \approx 6.63 \text{ V} \quad (1)$$

Use $R_{\text{top}} = 102 \text{ k}\Omega$, $R_{\text{bot}} = 14 \text{ k}\Omega$ (0.1% preferred for tighter setpoint).

2.3 5.0 V Logic & Peripheral Domain

2.3.1 Regulator

LMR14030S – 3 A asynchronous buck, HSOP-8-EP package. Common, inexpensive, well-stocked.

2.3.2 Load Budget (Worst-Case Simultaneous)

Load	Typical (mA)	Peak (mA)
Node2 ESP32-CAM (global camera, ESP-NOW TX burst)	200	400
Node3 ESP32-CAM (wrist camera, ESP-NOW TX burst)	200	400
Node1 ESP32-C6 offboard (leader board, UART + RF)	100	250
MIC5219-3.3 LDO input (feeds ESP32-S3 + TMC VIO)	400	620
Total 5.0 V rail	900	1 670

The LMR14030S 3 A rating provides $> 1.3\times$ headroom over worst-case peak. Inductor: 4.7 μH , $I_{\text{sat}} > 3.5$ A. Output: $\geq 100 \mu\text{F}$ SMD electrolytic + ceramic bank.

2.3.3 Node1 Leader Board Power Header (J_LDR)

Node1 (ESP32-C6 SuperMini, offboard) is powered from the 5.0 V rail via a 4-pin Molex Micro-Fit 3.0 connector (J_LDR) on the South Edge. A 10 μF ceramic bypass cap must be placed at the header pins. This header is *separate* from the UART signal connector (J_LDR); see Section 5.

2.4 3.3 V Logic Domain

Regulator: MIC5219-3.3YM5 (1 A, SOT-23-5). Replaces the AP2112K-3.3 (600 mA) to provide safe headroom. Same SOT-23-5 footprint; drop-in on schematic. Available from all major distributors at low cost.

Input capacitor: $\geq 10\ \mu\text{F}$ ceramic (0805) on +5V input pin.

Output capacitor: $22\ \mu\text{F}$ ceramic (0805) + $100\ \text{nF}$ X7R (0603) on +3.3V output pin.

Loads and peak current budget:

Load on 3.3 V rail	Peak (mA)
ESP32-S3-WROOM-1 core + RF	500
$4 \times$ TMC2209 V_{IO} pin (each $\approx 15\ \text{mA}$)	60
Pull-up resistor networks (10 nets)	20
Total	580

The MIC5219-3.3 1 A rating provides $\approx 420\ \text{mA}$ of headroom. Monitor LDO junction temperature (T_J) under sustained ESP-NOW burst transmit; worst-case dissipation $P_D = (5.0 - 3.3)\ \text{V} \times 0.58\ \text{A} \approx 1.0\ \text{W}$. Ensure adequate copper spreading under the SOT-23-5 pad.

3 4-Layer Substrate Stackup

All SMD components are restricted to **Layer 1 (Top)** for single-pass PCBA. Layer 4 carries no components; its copper is used for thermal spreading and signal escape only.

Layer	Type	Cu Weight	Thickness	Primary Function
L1 (Top)	Signal/Component	1.0 oz ($35\ \mu\text{m}$)	–	Component pads; $90\ \Omega$ USB differential pairs; buck switching node polygons; in-route decoupling.
L2	Ground Plane	1.0 oz ($35\ \mu\text{m}$)	–	Solid, unbroken GND. RF return shield. Thermal via landing plane for TMC2209 farms.
L3	Power Plane	1.0 oz ($35\ \mu\text{m}$)	–	Split power matrix: isolated copper polygons for +24V_MOTOR, +6.6V_SERVO, +5.0V, +3.3V.
L4 (Bottom)	Signal / Thermal	1.0 oz ($35\ \mu\text{m}$)	–	Trapped signal escape (STEP, DIR, SPI, DIAG). Large exposed copper pour under TMC2209 thermal via farms (no solder mask).

Table 1: Substrate Layer Architecture

Standard JLC4L stackup (or equivalent): L1–prepreg–L2–core–L3–prepreg–L4. Request controlled impedance: $90\ \Omega$ differential for USB pairs on L1.

4 Component Layout & Edge Zone Management

4.1 Perimeter Edge Allocations

4.1.1 North Edge – RF Keep-Out Zone

Reserved exclusively for the **ESP32-S3-WROOM-1-N16R8** module. The onboard PCB antenna must overhang the board edge by ≥ 2 mm. A copper keep-out zone across **all four layers** (15 mm $\times$ 10 mm) is mandatory directly below and around the antenna projection. **No connectors, mounting holes, vias, copper pours, or metallic enclosure features may occupy this zone.**

4.1.2 West Edge – Power & Communications

- **J_PWR** – XT30 right-angle connector, 24 V DC input / GND.
- **J_USB1** – USB-C receptacle, Native USB-OTG (ROS2 bridge / telemetry).
- **J_USB2** – USB-C receptacle, CH340K UART debug / firmware flash.

Power entry components (bulk electrolytic caps, TVS diode, reverse-polarity MOSFET if fitted) cluster immediately inboard of the West Edge to minimise high-current 24 V loop area.

4.1.3 South Edge – Actuators, Cameras, & Leader Link

All connectors use **vertical (straight)** orientation to eliminate right-angle horizontal clearance overhang.

- **J_M1 – J_M4** – Molex Micro-Fit 3.0, 4-pin, stepper motor phases (one per driver). May wrap to East Edge if South space is insufficient.
- **J_SV1 – J_SV3** – 3-pin 2.54 mm header, servo PWM / power / GND (Wrist Pitch, Wrist Yaw, Gripper).
- **J_CAM1** – 2-pin Molex Micro-Fit 3.0, 5 V / GND for Node2 (global perspective ESP32-CAM).
- **J_CAM2** – 2-pin Molex Micro-Fit 3.0, 5 V / GND for Node3 (wrist / eye-in-hand ESP32-CAM).
- **J_LDR** – 4-pin 2.54 mm header, Leader UART signals (TX, RX, GND, 5 V) to Node1 ESP32-C6.

4.1.4 East Edge – Overflow Stepper Headers

If stepper headers J_M3 and J_M4 cannot fit on the South Edge alongside the servo and camera connectors, they relocate to the East Edge in the same vertical orientation.

4.1.5 Interior Zones

Power Domain (West-Centre): 6.6 V and 5.0 V bucks, MIC5219-3.3 LDO, all bulk electrolytic capacitors. Clustered adjacent to the West Edge.

Driver Matrix (East-Centre): 4 $\times$ TMC2209 QFN-28 in a 2 $\times$ 2 grid. Thermal via farms mandatory. Motor phase traces route directly outward to edge headers.

Logic & Memory Core (Centre-South): MicroSD slot and critical decoupling passives. Maximises isolation from buck switching nodes.

5 Full Connector Reference

Ref	Type	Signal / Net	Edge	Power Rail
J_PWR	XT30 RA 2-pin	24 V DC in / GND	West	24 V
J_USB1	USB-C receptacle	Native USB-OTG (D ⁻ /D ⁺)	West	–
J_USB2	USB-C receptacle	CH340K UART debug/flash	West	–
J_M1	Micro-Fit 3.0 4-pin	J1 Base stepper phases A1/A2/B1/B2	South/East	24 V
J_M2	Micro-Fit 3.0 4-pin	J2 Shoulder A phases	South/East	24 V
J_M3	Micro-Fit 3.0 4-pin	J3 Shoulder B phases	South/East	24 V
J_M4	Micro-Fit 3.0 4-pin	J4 Elbow phases	South/East	24 V
J_SV1	2.54 mm 3-pin	Wrist Pitch PWM / 6.6 V / GND	South	6.6 V
J_SV2	2.54 mm 3-pin	Wrist Yaw PWM / 6.6 V / GND	South	6.6 V
J_SV3	2.54 mm 3-pin	Gripper PWM / 6.6 V / GND	South	6.6 V
J_CAM1	Micro-Fit 3.0 2-pin	Node2 5 V / GND	South	5.0 V
J_CAM2	Micro-Fit 3.0 2-pin	Node3 5 V / GND	South	5.0 V
J_LDR	2.54 mm 4-pin	Node1 UART TX, RX, GND, 5 V	South	–
J_SD	MicroSD push-pull	SPI: CS / MOSI / SCK / MISO	Interior	3.3 V

Table 2: Complete Connector Directory

6 Active Component Directory (KiCad Symbol & Footprint)

Component	KiCad Symbol	KiCad Footprint
ESP32-S3-WROOM-1-N16R8	MCU_Espressif:ESP32-S3-WROOM-1	RF_Module:ESP32-S3-WROOM-1
4× TMC2209-LA	Driver_Motor:TMC2209	Package_DFN_QFN:QFN-28-1EP_5x5mm_P0.5mm
TPS54540B (6.6 V buck)	Regulator_Switching:TPS54540B	Package_S0:HSOP-8-1EP_3.89x4.89mm_P1.27mm
LMR14030S (5.0 V buck)	Regulator_Switching:LMR14030S	Package_S0:HSOP-8-1EP_3.89x4.89mm_P1.27mm
MIC5219-3.3YM5 (LDO)	Regulator_Linear:MIC5219-3.3YM5	Package_T0_SOT_SMD:SOT-23-5
CH340K UART Bridge	Interface_USB:CH340K	Package_S0:ESSOP-10_3.9x4.9mm_P1.0mm
MicroSD Card Holder	Connector:MicroSD_Card_Kit	Connector_Card:MicroSD_Hirose_DM3D-SF

Table 3: Symbol and Footprint Directory

7 GPIO Allocation & Net Assignment

7.1 Locked / Forbidden GPIO Lines

The following GPIOs are consumed by the N16R8 internal silicon and must not appear on any net, pad, pull resistor, or via on this PCB:

GPIO	Reason
GPIO26	Internal octal flash clock
GPIO33	Flash D4
GPIO34	Flash D5
GPIO35	Octal PSRAM D6 / WP
GPIO36	Octal PSRAM D7 / HOLD
GPIO37	Octal PSRAM CLK

7.2 Boot Strapping

- GPIO0 → MCU_BOOT: 10 k Ω pull-up to 3.3 V. Tactile switch to GND for manual boot-mode override.
- GPIO46 → MCU_STRAP_LOW: 10 k Ω pull-down to GND. Required by ESP32-S3 strapping specification.
- GPIO45: Leave floating (internal OTP default).

7.3 Full GPIO Allocation Table

GPIO	Net Name	Notes
19	USB_D-	90 Ω diff pair → J_USB1. Route on L1 only.
20	USB_D+	90 Ω diff pair → J_USB1. Route on L1 only.
43	UART0_TXD	UART0 TX → CH340K RXD (Pin 3).
44	UART0_RXD	UART0 RX ← CH340K TXD (Pin 4).
10	SD_CS	SPI chip select; 10 k Ω pull-up to 3.3 V.
11	SD_MOSI	SPI MOSI; 10 k Ω pull-up to 3.3 V.
12	SD_SCK	SPI clock.
13	SD_MISO	SPI MISO; 10 k Ω pull-up to 3.3 V.
4	DRV_EN_ALL	Active-LOW; shared EN bus → Pin 18 of all 4× TMC2209.
5	J1_BASE_STEP	→ TMC2209-1 Pin 16 (STEP).
6	J1_BASE_DIR	→ TMC2209-1 Pin 17 (DIR).
7	J1_BASE_PDN	Single-wire UART. 1 k Ω series + 4.7 k Ω pull-up to 3.3 V. → TMC2209-1 Pin 9 (PDN_UART).
1	J1_BASE_DIAG	StallGuard / fault flag. 47 k Ω pull-up to 3.3 V. ← TMC2209-1 Pin 26 (DIAG).
8	J2_SHLDA_STEP	→ TMC2209-2 Pin 16 (STEP).
9	J2_SHLDA_DIR	→ TMC2209-2 Pin 17 (DIR).
14	J2_SHLDA_PDN	Single-wire UART. 1 k Ω series + 4.7 k Ω pull-up to 3.3 V. → TMC2209-2 Pin 9.
2	J2_SHLDA_DIAG	StallGuard / fault flag. 47 k Ω pull-up to 3.3 V. ← TMC2209-2 Pin 26 (DIAG).
15	J3_SHLDB_STEP	→ TMC2209-3 Pin 16 (STEP).
16	J3_SHLDB_DIR	→ TMC2209-3 Pin 17 (DIR).
17	J3_SHLDB_PDN	Single-wire UART. 1 k Ω series + 4.7 k Ω pull-up to 3.3 V. → TMC2209-3 Pin 9.
3	J3_SHLDB_DIAG	StallGuard / fault flag. 47 k Ω pull-up to 3.3 V. ← TMC2209-3 Pin 26 (DIAG).
18	J4_ELBOW_STEP	→ TMC2209-4 Pin 16 (STEP).
21	J4_ELBOW_DIR	→ TMC2209-4 Pin 17 (DIR).
47	J4_ELBOW_PDN	Single-wire UART. 1 k Ω series + 4.7 k Ω pull-up to 3.3 V. → TMC2209-4 Pin 9.
48	J4_ELBOW_DIAG	StallGuard / fault flag. 47 k Ω pull-up to 3.3 V. ← TMC2209-4 Pin 26 (DIAG).

continued...

GPIO	Net Name	Notes
38	SRV_WRIST_PITCH	LEDC/MCPWM PWM $\rightarrow$ J_SV1 signal pin.
39	SRV_WRIST_YAW	LEDC/MCPWM PWM $\rightarrow$ J_SV2 signal pin.
40	SRV_GRIPPER	LEDC/MCPWM PWM $\rightarrow$ J_SV3 signal pin.
41	LDR_UART_TX	921 600 baud, 8N1 $\rightarrow$ Node1 ESP32-C6 RX (J_LDR Pin 1).
42	LDR_UART_RX	921 600 baud, 8N1 $\leftarrow$ Node1 ESP32-C6 TX (J_LDR Pin 2).
0	MCU_BOOT	10 k Ω pull-up; tactile switch to GND.

8 Passive Decoupling & Filtering Networks

8.1 TMC2209 Per-Driver Filtering (Repeat $\times 4$)

All capacitors placed within 2 mm of IC package edge. Reference TMC2209 datasheet application circuit throughout.

- **VM bulk:** 100 μ F / 35 V low-ESR SMD electrolytic (Panasonic FR / Nichicon UWX, case D). **Electrolytic only – no tantalum.** Between +24V_MOTOR and GND_D.
- **VM HF bypass:** 100 nF X7R 50 V (0603). Trace to Pin 27 (VM) $\leq$ 1 mm.
- **VIO filter:** 1.0 μ F X7R (0603) from +3.3V to Pin 12 (VIO).
- **Charge pump:** 100 nF X7R 50 V (0603) between Pin 22 (VCP) and Pin 24 (CPI).
- **Internal regulator:** 4.7 μ F X5R (0603) from Pin 25 (VOUT) to GND.
- **DIAG pull-up:** 47 k Ω (0402) from Pin 26 (DIAG) to +3.3V. DIAG is open-drain; pull-up is mandatory for valid logic level.

8.2 CH340K Decoupling

- 100 nF X7R 10 V (0402) on VCC supply pin to GND, placed $\leq$ 1 mm from IC.
- 100 nF X7R 10 V (0402) on V3 (internal 3.3 V reference pin) to GND.
- 100 nF X7R (0402) on XI USB signal conditioning pin per CH340K datasheet recommendation.

8.3 ESP32-S3-WROOM-1 Module Decoupling

Per Espressif hardware design guidelines for the WROOM-1 module:

- 10 μ F + 100 nF ceramic on the 3V3 module supply pin cluster to GND, placed directly adjacent to the module pad row.
- Place decoupling capacitors *in-route*: power plane via $\rightarrow$ capacitor pad $\rightarrow$ IC supply pin. Do not branch off a parallel stub.

8.4 MicroSD Card Slot

- 10 k Ω pull-up resistors on SD_CS, SD_MOSI, SD_MISO to +3.3V.

- 100 nF bypass cap on VCC supply rail of card slot to GND.

8.5 Node1 / Node2 / Node3 Power Header Bypass

- 10 μ F ceramic cap on 5 V at each of J_CAM1, J_CAM2, J_LDR_PWR headers to absorb ESP-NOW TX current burst ($\approx$ 200 mA step).

9 High-Density Layout & Thermal Rules

9.1 In-Route Decoupling Rule

Decoupling capacitors must be placed *in-series* with the power trace, not tapped off a stub:

Power Plane Via $\rightarrow$ Cap pad (0402/0603) $\rightarrow$ IC supply pin

This eliminates the inductance of a parallel stub branch. Applies to all TMC2209 VM caps, all 3.3 V / 5 V decoupling on the ESP32-S3 module pads, and the CH340K VCC cap.

9.2 TMC2209 Thermal Via Farms

Each TMC2209 QFN-28 must dump its entire thermal load through the PCB substrate.

1. Place a 3×3 via farm inside the exposed centre ground pad (EP) of each driver footprint.
2. Drill diameter: 0.3 mm. Grid pitch: $\leq$ 1.0 mm.
3. Vias must connect L1 $\rightarrow$ L2 (solid GND plane) and extend to L4.
4. On Layer 4, draw a solid copper pour matching the TMC2209 driver grid footprint area with no solder mask (B.Mask cutout in KiCad).
5. This converts the PCB bottom surface into a passive thermal radiator.

9.3 Buck Switching Node Isolation

The SW nodes of both buck converters, their freewheeling diodes, and inductor pads must be routed as ultra-wide, short copper polygons on Layer 1.

No signal trace on L1 or L4 may pass beneath these switching node polygons.

Affected critical nets: ESP-NOW RF path (Node2/Node3 timing), SPI MicroSD transactions, Leader UART receive path (LDR_UART_RX).

9.4 Ground Stitching

Place GND stitching vias (L1 $\rightarrow$ L2):

- Board perimeter: every $\approx$ 5 mm.
- Along the full length of both USB differential pair routes.
- Immediately adjacent to all buck regulator switching polygons.
- Between each power domain zone boundary on L3.

Servo high-current GND return must not share any trace segment with MCU signal GND. Both must terminate at the L2 GND plane via *separate* vias before any common-mode coupling path can form.

10 Net Class Trace Width & Via Rules

Enter the following parameters into the KiCad **Board Setup** → **Net Classes** manager.

Net Class	Member Nets	Min Width	Min Gap	Via (drill/pad)
Power_24V	+24V, +24V_MOTOR	1.20 mm	0.40 mm	0.4/0.7 mm stitched
Power_Servo	+6.6V_SERVO	1.50 mm	0.30 mm	0.4/0.7 mm stitched
Power_Logic	+5.0V, +3.3V	0.50 mm	0.20 mm	0.3/0.6 mm standard
Motor_Phase	J1_A1...J4_B2	0.80 mm	0.35 mm	0.4/0.7 mm standard
USB_Diff	USB_D+, USB_D-	0.25 mm	0.20 mm	0.25/0.5 mm micro
Default_Sig	STEP, DIR, SPI...	0.15 mm	0.15 mm	0.3/0.6 mm standard

Table 5: Net Class Routing Parameters

11 Peripheral Interface Notes

11.1 Wireless Camera Nodes (Node2 / Node3 via ESP-NOW)

J_CAM1 and J_CAM2 supply **5 V power and GND only**. All data is exchanged wirelessly:

- Protocol: ESP-NOW, channel 6, stateless packet send.
- Payload: 128-byte brightness histogram feature vector embedded in `ExperiencePacket` (magic 0x4E4D5650).
- Freshness gate: Node0 discards frames older than 300 ms before IK merge.

11.2 Leader UART – Node1 ESP32-C6 (Offboard)

Node1 is an ESP32-C6 SuperMini board connected via two separate connectors:

- **J_LDR_PWR**: 5 V / GND power (Micro-Fit 3.0, 2-pin, South Edge).
- **J_LDR**: UART signals TX / RX / GND / spare (2.54 mm 4-pin, South Edge).

Baud: 921 600, 8N1, non-blocking async receive. Parser uses magic-number scan + CRC-CCITT (poly 0x1021) before packet consumption. Packets carry absolute joint target values for all 6 joints.

11.3 Dual USB-C Interface

Port 1 – J_USB1: Native USB-OTG. 90 Ω differential pair directly on GPIO19 (D-) / GPIO20 (D+). Used for ROS2 USB-CDC bridge, telemetry streaming, VLA dataset logging.

Port 2 – J_USB2: CH340K UART bridge. GPIO43 (TXD) / GPIO44 (RXD). DTR/RTS auto-reset for PlatformIO upload. Eliminates need for external USB-to-serial programmer during development.

11.4 MicroSD Card

Hardware SPI. FAT32, 32 kB clusters. 1 MB pre-allocated circular buffer for experience packets. SD flush every 8 writes to reduce blocking. Pull-ups: 10 k Ω on MOSI, MISO, CS lines.

12 Revision History

Rev	Change	Detail
Alpha Proto	Baseline	Through-hole DevKit carrier; TMC2209 modules; single-layer protoboard.
Beta Rev1	First custom PCB	60×60 mm 4-layer; TMC2209 QFN direct IC; dual USB-C; single-sided SMT; AP2112K-3.3 LDO (600 mA).
Beta Rev2	This document	LDO upgraded to MIC5219-3.3 (1 A); TMC2209 DIAG pins routed (GPIO1/2/3/48) for StallGuard2 / CoolStep; Bulk caps changed to low-ESR SMD electrolytic (tantalum removed); 5 V load budget revised for Node1 ESP32-C6 off-board; Node1 split into J_LDR (UART) + J_LDR_PWR (5 V power) connectors; ESP32-CAM headers changed to 2-pin Micro-Fit 3.0; Module variant locked to N16R8 (octal PSRAM); GPIO35/36/37/33/34/26 exclusions enforced; CH340K decoupling added (3× 100 nF); Full connector edge-zone map added to Section 4 and connector table in Section 5.

Table 6: Revision History

Firmware note: `board.config.h` and `tmc2209.config.h` must be updated to reflect the Rev 2 GPIO mapping (specifically DIAG pin additions on GPIO1, 2, 3, 48) before flashing to Beta Rev 2 hardware.